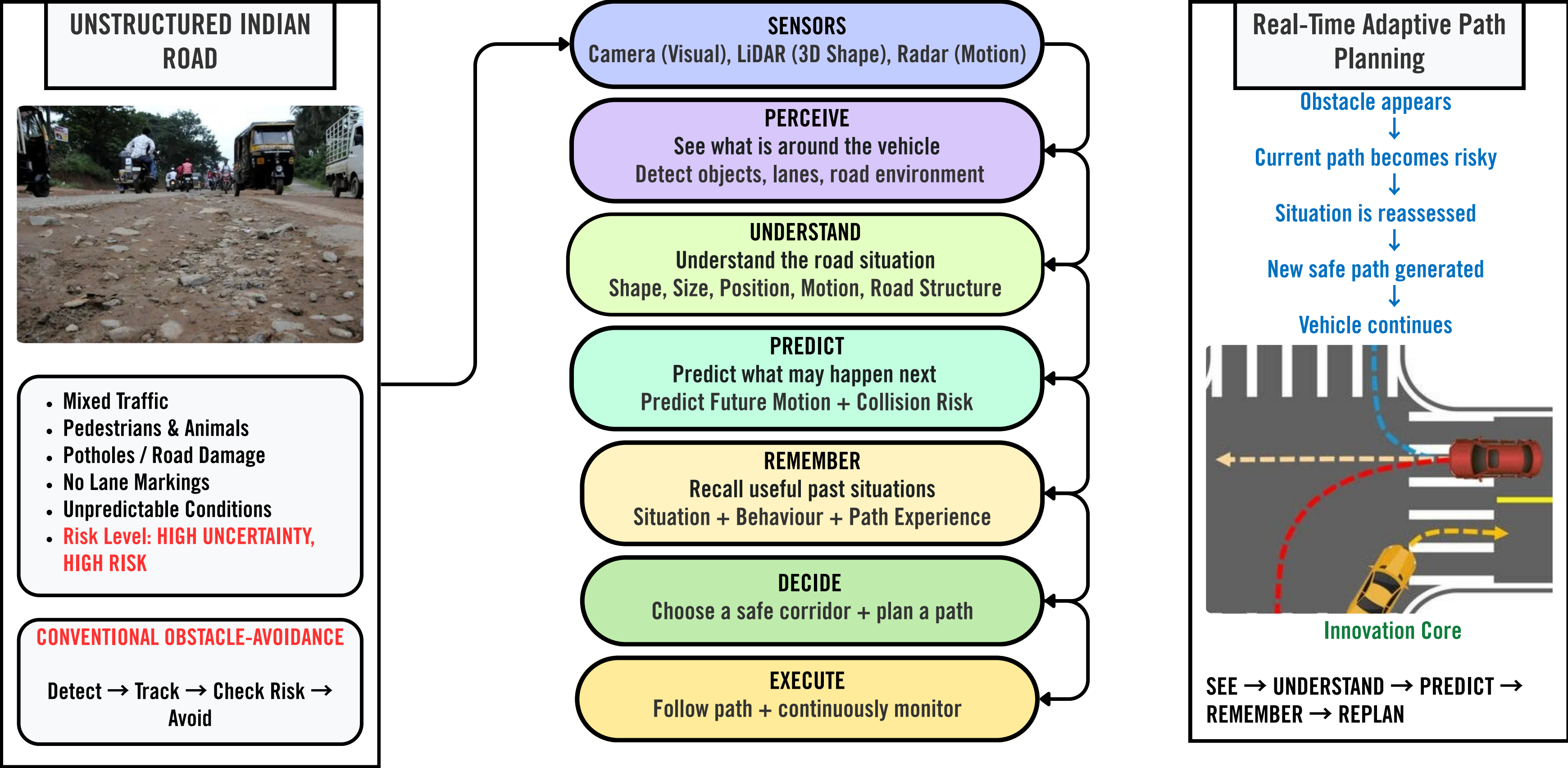


TITLE PAGE

- Problem Statement ID – 26037
- Problem Statement Title- Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads
- Theme- Smart Vehicles
- PS Category- Software
- Team ID- 129107
- Team Name - ALTHEA_AURORA



“The vehicle reasons about the road situation — not just the object.”



TECHNICAL APPROACH

14-Phase End-to-End Pipeline | Technologies | Methodology | Implementation

TECH STACK

**AI / Logic**

- Python 3.10
- NumPy
- OpenCV

**Real-Time / Deployment**

- C++
- Lightweight algorithms

**Simulation**

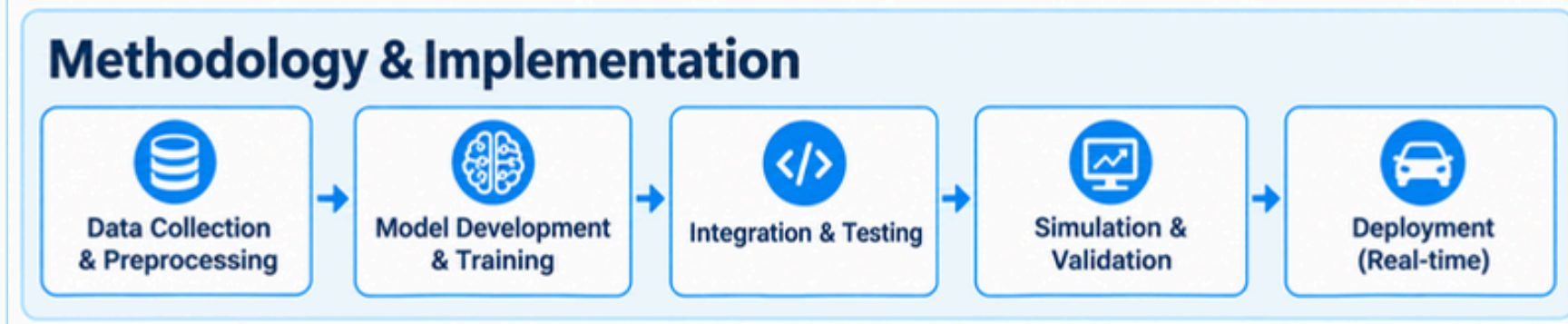
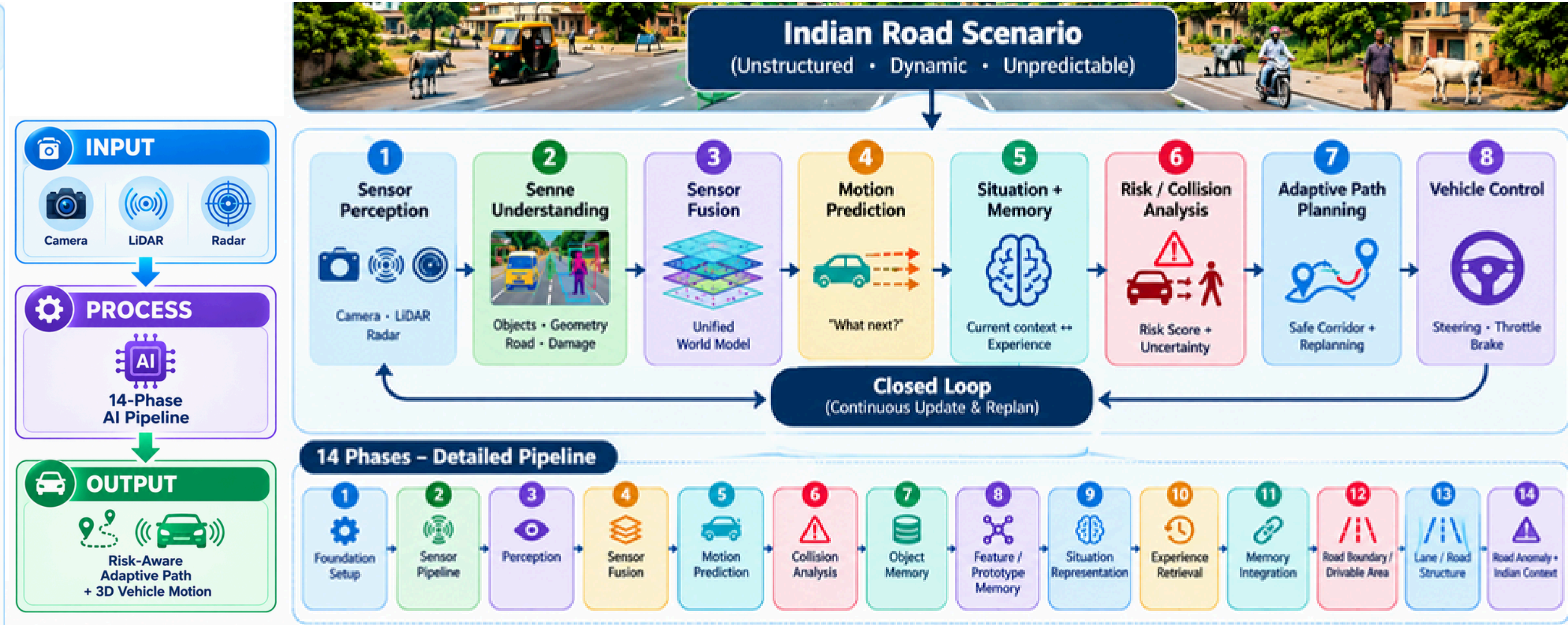
- Three.js
- 3D Scene
- Vehicle Motion

**Sensors**

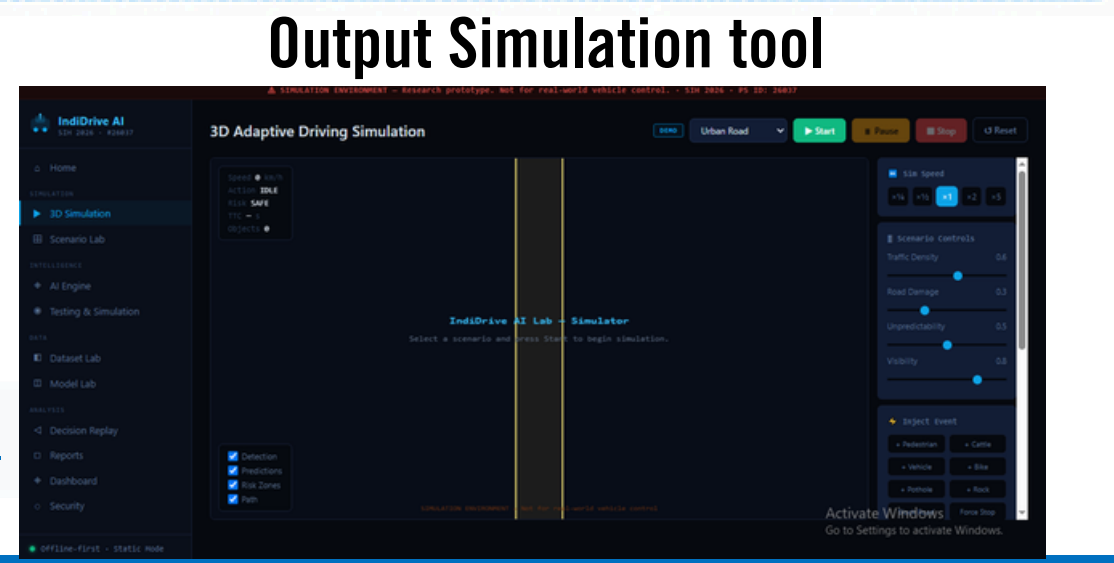
- Camera
- LiDAR
- Radar

**AI Memory**

- Structured Features
- Prototypes
- Experience Retrieval



Project Development: <https://github.com/ARULGNANAKUMAR/indidrive-ai-development.git>



**WHY IT IS FEASIBLE**

01

**Modular Architecture**
Perception → Fusion → Prediction → Planning → Control

02

**Offline-First**
Core decision pipeline does not depend on continuous internet connectivity.

03

**Simulation-Ready**
Scenarios can be tested repeatedly before real-world deployment.

04

**Incremental Deployment**
Python (research & development) → C++ (real-time implementation)



 KEY CHALLENGES & RISKS		 OUR STRATEGY	
	Missing lane markings		Road-boundary + drivable-area understanding
	Mixed traffic (autos, bikes, buses, etc.)		Multi-object tracking + situation representation
	Sudden pedestrian / cattle movement		Short-term motion prediction + risk analysis
	Potholes / gravel / waterlogging		Road anomaly understanding
	Dynamic obstacle appears		Continuous adaptive replanning
	Sensor uncertainty / partial sensor loss		Camera + LiDAR + Radar fusion

Risk Firewall: "Uncertainty is not ignored — it is continuously measured, predicted and handled."

UNCERTAINTY → SENSOR FUSION → PREDICTION → RISK ANALYSIS → SAFE REPLANNING → SAFE MOTION

IMPACT AND BENEFITS

ONE ADAPTIVE DECISION → MULTIPLE IMPACTS

1

SOCIAL IMPACT

Potential for safer and more inclusive mobility

 Pedestrians

 Two-wheelers

 Cattle / Road users

 Safer interaction with autonomous vehicle

✓ Handles mixed & unpredictable road users

✓ Supports safer movement in dense / underserved environments

✓ Considers pedestrians, animals and diverse vehicles in planning

2

MOBILITY / OPERATIONAL BENEFIT

Continuous and adaptive movement

 Unexpected obstacle

 Risk assessment

 Path replanning

 Continuous movement

✓ Adaptive path instead of fixed-path behaviour

✓ Real-time response to changing road situations

✓ Supports village roads, urban intersections, markets and highways

3

ECONOMIC / SYSTEM BENEFIT

Efficient, scalable and cost-conscious approach

 Structured perception

 Prediction

 Experience reuse

 Efficient autonomous decision-making

✓ Reusable structured experience instead of repeatedly reasoning from raw scene data

✓ Modular architecture supports incremental development

✓ Simulation-based validation can reduce dependence on early physical-road testing

4

POTENTIAL EFFICIENCY BENEFIT

More efficient movement, lower overall footprint

 Obstacle detected

 Early decision

 Smooth path adaptation

 Potentially less unnecessary stopping / manoeuvring

✓ Smooth, situation-aware motion planning can support more efficient vehicle movement.

✓ May help reduce unnecessary idling and energy waste.

✓ Contributes to cleaner, smarter, and more sustainable mobility.



“Make autonomous driving adapt to Indian roads — not force Indian roads to behave predictably.”

Autonomous Driving Research

- Multi-sensor perception & fusion
- Object detection, tracking & scene understanding
- Motion prediction for dynamic road users
- Reference: IEEE — Multi-Sensor Fusion in Automated Driving: A Survey

Link: [IEEE Research Paper](#)

Motion Planning Research

- Sensor-fusion based motion planning
 - Future waypoint / trajectory prediction
 - Dynamic obstacle-aware navigation
 - Reference: IEEE — An End-to-End Motion Planner Using Sensor Fusion for Autonomous Driving
- Link: [IEEE Research Paper](#)

MathWorks Implementation Research

- Automated Driving Toolbox
- Detection, tracking & sensor fusion
- Path planning & vehicle control
- MATLAB / Simulink / RoadRunner workflows

Links:

[Automated Driving Toolbox](#)

[Automated Driving Algorithms](#)

Simulation + Project Resources

- RoadRunner 3D scenario simulation
- Sensor-data generation & scenario testing
- Closed-loop planning / control validation

Links

GitHub (Project Development):

[\[https://github.com/ARULGNANAKUMAR/indidrive-ai-development.git\]](https://github.com/ARULGNANAKUMAR/indidrive-ai-development.git)

Demo Simulation tool : [\[https://indidrive-ai-prototype.vercel.app/\]](https://indidrive-ai-prototype.vercel.app/)



“Research references were studied to guide perception, sensor fusion, prediction, planning and simulation design.”